

Justification and Approval (J&A)

“Source Selection Information – see FAR 2.101 and 3.104”

1. Identification of the agency and contracting activity.

a. Federal agency and contracting activity.

Department of Health and Human Services/National Institutes of Health

b. Sponsoring organization.

HHS/NIH/NIAID/DMID

c. Project Officer information.

- **Project Officer name:** Jeremy Starr
- **Mailing address:** 5601 Fishers Lane, Bethesda MD, 20892
- **E-mail address.** Jeremy.Starr@nih.gov
- **Telephone number.** 240-268-4711

2. Nature and/or description of the action being approved.

- a. Acquisition purpose and objectives.** The purpose of the acquisition is to continue the work currently performed under 75N93021C00002 – Chemistry Center for Combating Antibiotic Resistant Bacteria – for an additional five years of active chemistry and three years of compound management. The driving initiative is the Combatting Antibiotic Resistant Bacteria (CARB) National Action Plan (<https://www.hhs.gov/sites/default/files/carb-national-action-plan-2020-2025.pdf>) of which CC4CARB serves a vital function in addressing the need for new chemical matter purpose built for penetrating and killing Gram-negative bacteria – the majority of pathogen multidrug resistant organisms.
- b. Project background.** The project is a renewal/continuation of contract 75N93021C00002 (Chemistry Center for Combating Antibiotic Resistant Bacteria) which was awarded in open competition. The contractor is RTI International, and the anticipated period of performance is 02/01/2021 to 01/31/2026. Total obligated so far is \$15,164,408. The contract Project background: Multidrug-resistant (MDR) bacteria are key pathogens responsible for the alarming rates of drug resistance in healthcare and community settings. The Centers for Disease Control and Prevention (CDC) has developed a list of organisms considered Antimicrobial

Resistance (AMR) threats. Many of these organisms are MDR Gram-negative bacteria, including carbapenem-resistant Enterobacteriaceae, MDR Acinetobacter and MDR Pseudomonas aeruginosa. Resistance has evolved over the past decade to the point that for some infections there are no effective treatment options. The problem is exacerbated by the scarcity of novel compounds effective against the modern cohort of drug resistant bacteria and the current antibacterial drug discovery pipeline is inadequate to the need.

In publications and public fora (such as NIAID workshops), the antibacterial research community has identified the lack of suitable compound screening libraries and novel chemical scaffolds as a significant bottleneck for the discovery and development of novel antibacterial drugs. In particular, there is a scarcity of novel chemical classes and natural compounds with properties appropriate for Gram-negative cell penetration (and efflux avoidance). Most public and privately held compound libraries are enriched with compounds previously designed for human targets and pathways and, thus, they are poorly suited as starting points for discovery of drugs that should avoid human targets and pathways, and which require properties for bacterial cell penetration rather than mammalian cell penetration. In a report produced with the help of key opinion leaders by the PEW Charitable Trust titled “Scientific Roadmap for Antibiotic Discovery,” the lack of appropriate chemical libraries is recognized as one of the key problems in antibacterial drug discovery. This report states that the need to “generate and tailor chemical matter for antibacterial discovery” and to “build prototype libraries tailored for antibacterial discovery” is a fundamental part of the critical path to new antibacterial agents. Only recently have rules emerged that describe chemical space for improved antibacterial activity, though these are still neither comprehensive nor fully understood. We expect continued focused academic and industrial research to expand and refine this nascent understanding and we anticipate CC4CARB will leverage it in the design, synthesis and in vitro testing of new antibacterial compounds.

In 2021, NIAID awarded the first CC4CARB contract (75N93021C00002) to Research Triangle Institute (RTI). Under that contract an innovative and one-of-a-kind chemistry center was established to synthesize and deliver rationally designed compound libraries to the scientific community for use in antibacterial drug discovery programs, particularly those targeting Gram-negative bacteria. This center fits within the broader aims put forth in the National Action Plan for Combating Antibiotic-Resistant Bacteria (CARB) by synthesizing and delivering new compounds that enable an understanding of SAR for optimal activity against antibacterial resistant pathogens and that provide the basis for a robust antibacterial development pipeline. Critical to the success of this center have been three core requirements: Library Design, Compound Synthesis and Compound Management.

This Request For Proposals (RFP) contemplates a continuation of the current CC4CARB initiative with additions and updates to reflect lessons learned and with the intent of building on and expanding, rather than revisioning or replacing, the successful model established by the contract awarded in 2021. Infections caused by antibiotic resistant bacteria are an ongoing global health problem that will need new chemistry approaches for a long time to come. The true need far exceeds the current scale of CC4CARB. When the current CC4CARB began to address this need and gain public acceptance and visibility, the level of interest from external researchers exceeded what could be completed with the original planned resources of CC4CARB. The extension of CC4CARB helps to address this need and maintain NIAID involvement at the forefront of discovering new chemical entities to address antibiotic resistance.

3. Description of the supplies or services required to meet the agency's needs (including the estimated value).

- a. Project title.** Chemistry Center for Combating Antibiotic Resistant Bacteria (CC4CARB)
- b. Project description.** CC4CARB is a chemistry center where external antibacterial researchers collaborate with CC4CARB chemists to produce new fit-for-purpose chemical libraries targeting primarily Gram-negative pathogens. All new compounds are to be tested for in vitro potency and made available to the public free of charge as starting points for new drug discovery programs. External outreach and community engagement is a major component of the project which depends for its success on the long term relationships the center builds with its collaborators. For additional detail, the SOW is provided as an Appendix at the end of this document.

• **Requirement type.**

- ☒ Research & development (R&D)
☐ R & D support services
☐ Support services (non-R&D)
☐ Supplies/equipment
☐ Information technology (IT)
☐ Construction
☐ Architect-engineer (A & E) services
☐ Design-build
☐ Other (specify): _____

• **Type of action.**

- ☐ New requirement
☒ Follow-on
☐ Other (specify): _____

• **Proposed contract/order type.**

- ☐ Firm-fixed-price
☐ Other fixed-price (specify, e.g., fixed-price award-fee, fixed-price incentive-fee): _____
☒ Cost-plus-fixed-fee
☐ Other cost reimbursement (specify, e.g., cost-plus-award-fee, cost-plus-incentive-fee): _____
☐ Time and materials
☐ Indefinite delivery (specify whether indefinite quantity, definite quantity, or requirements): _____

- ☒ Other (specify): Cost Plus Fixed Fee Level of Effort
☐ Completion Form ☒ Term form

- **Acquisition identification number.**

RFP No. 75N93024R00022

- c. Total estimated dollar value and performance/delivery period.** The estimated new cost for the first 5 years is \$25,000,000 and \$3,000,000 for the final three years for a total 8 years and \$28,000,000. These costs are in addition to already obligated funds under the current CC4CARB contract.

4. Identification of the statutory authority permitting other than full and open competition.

☒ This acquisition is conducted under the authority of 41 United States Code (U.S.C.) 253(c)(3)(B) as set forth in Federal Acquisition Regulation FAR 6.302-1(b)(1)(i)

☐ This acquisition is conducted under the authority of section 4202 of the Clinger-Cohen Act of 1996.

☐ This acquisition is conducted under the authority of the Services Acquisition Reform Act of 2003 (41 U.S.C. 428a).

5. Demonstration that the proposed contractor(s) unique qualifications or the nature of the acquisition requires use of the authority cited.

- a. Name and address of the proposed contractor(s).** The proposed sole source contractor is RTI International at 3040 East Cornwallis Road P.O. Box 12194, Research Triangle Park, NC 27709-2194

- b. Nature of the acquisition and proposed unique qualifications of the contractor(s).**

CC4CARB is a complex contract involving multiple functions working together to deliver a valuable and unique service to the public. CC4CARB provides medicinal chemistry design, synthesis, compound management, and biological activity testing of new antibacterial agents. It also serves as a repository of chemical samples specifically aimed at multi-drug-resistant bacteria available for screening activities. It performs these services through a network of collaborations with antibacterial researchers in academia and private industry, worldwide. CC4CARB depends on an integrated outreach function that RTI has painstakingly grown, and continues to build, a broad network of relationships with antibacterial researchers who trust CC4CARB and its leadership to be thoughtful stewards of their intellectual property and scientific programs in the conduct of CC4CARB activities. All of the infrastructure for outreach, chemistry, compound management, and the immensely valuable network of relationships with collaborators is integrated with the current

contractor's institution and would be lost, or incur significant delays and cost, if transitioned to another contractor. No other organization worldwide is currently performing this work due to the large cost and low rate of return. In absence of Government funding, this program would not exist anywhere.

It is estimated that CC4CARB would incur \$3.6M of preventable costs and 9-12 months of preventable delays if awarded to a new contractor. In addition to these costs, the transition would disrupt the relationships and trust built between CC4CARB and collaborators, many of whom may not follow CC4CARB to its new location and endure the long delays of rebuilding the program in the new location. These relationships, which cannot be assigned a dollar or time value, per se, are essential for the success of CC4CARB and will be irreversibly diminished by moving CC4CARB to another contractor. For the reasons stated, the incumbent is the only vendor which possesses the unique capability to keep the CC4CARB activities running without incurring a critical loss in services, data, and programmatic infrastructure.

Therefore, this request is citing the authority in FAR 6.302-1 (b)(1)(i) – Unique supplies or services available from only one source or only one supplier with unique capabilities.

6. **Description of the efforts made to ensure that offers are solicited from as many potential sources as practicable. Indicate whether a FedBizOpps notice was or will be publicized as required by FAR Subpart 5.2 and, if not, which exception under FAR 5.202 applies.** The authority cited is FAR 6.302-1 and there are no other contractors who can continue the work without significant disruption. A Sources Sought Notice was published on June 17, 2024, seeking capability statements and six (6) capability statements were received from small business organizations. A presolicitation Notice was published on October 17, 2024 to give interested parties advance notice about the upcoming solicitation. The presolicitation notice closed on November 1, 2024, with no objections.
7. **Determination by the Contracting Officer that the anticipated cost/price to the Government will be fair and reasonable.** Upon entering negotiations with RTI, the Contracting Officer will use appropriate and available information to review proposed prices and determine that they are fair and reasonable, and such documentation shall be included in the contract file. The contracting officer will review the offerors' current contract for areas where cost savings and efficiencies can be realized. Individual cost elements are going to be analyzed to determine realism. In addition, other information that may be used to determine fair and reasonableness includes Independent Government Cost Estimates (IGCE), historical information, including actual costs of similar chemistry conducted under NIAID preclinical services (Interventional Agents Task Areas B and C), and commercial pricing. Closest analogues to the acquired services will be reviewed to determine the fairness of the offeror's proposed price. Actions taken under the authority of this justification shall include a documentation of the price/cost analysis conducted and whether the proposed price is fair and reasonable.

8. Description of the market research conducted (see FAR Part 10) and the results, or a statement of the reasons market research was not conducted.

On June 17, 2024, a Small Business Sources Sought Announcement seeking capability statements was issued through the Government wide point of entry, System for Award Management (SAM), with a response deadline of July 5, 2024. NIAID received six (6) of capability statements from small business organizations. The capability statements were reviewed for demonstrated capability of the respondents to deliver the Government requirement as described in the notice as including access to facilities, personnel and project management expertise at the scale and duration required to complete the Statement of Work (SOW) at low risk to the Government. Upon review, it was determined that none of the capability statements submitted adequately met the requirements outlined in the Statement of Work (SOW).

CC4CARB is a complex contract involving multiple functions working together to deliver a valuable and unique service to the public. Other businesses lack the necessary network, resources, infrastructure, and capabilities to keep the programmatic activities ongoing. Therefore, the incumbent is the only source capable of sustaining this unique program without a critical loss in services, which NIAID is unlikely to recover. Notably, the incumbent has devoted significant resources and built a network and infrastructure to maintain this unprecedented program without critical programmatic losses.

As a result of Market Research, the Contracting Officer determined that it is in the Government's best interest to sole source these unique and complex requirements to the incumbent contractor.

Duplicative costs are shown below:

Impact	Duplicated Cost	Time delay
Rebuilding website and Library Idea management infrastructure	\$250,000	6 months
Rebuilding relationships with collaborators – costs of duplicated labor and advertising	\$500,000	1 year or more
Rebuilding the scientific advisory board – cost of duplicated labor	\$100,000	3 months
Transitioning the existing library – labor costs plus lost inventory	\$250,000 plus \$250,000 in lost compound value	6 months
Productivity delays at \$3000 per compound	750 compounds = \$2,250,000	9 months
Total preventable cost of award to another contractor	\$3,600,000	9-12 months

9. Any other facts supporting the use of other than full and open competition. Not applicable.

10. Listing of sources, if any, that expressed, in writing, an interest in the acquisition. In response to a Small Business Sources Sought notice published on June 17, 2024, NIAID received six (6) of capability statements from small business organizations. The capability statements were reviewed for demonstrated capability of the respondents to deliver the Government requirement as described in the notice as including access to facilities, personnel and project management expertise at the scale and duration required to complete the Statement of Work (SOW) at low risk to the Government. Upon review, it was determined that none of the capability statements submitted adequately met the requirements outlined in the Statement of Work (SOW).

11. Statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the required supplies or services. There are no future plans to acquire this requirement subsequent to the current solicitation. A transition plan is required as part of this effort and is intended to reduce barriers to any and all future work, should this program be needed. The transition plan will only be implemented in the event of selection of a new contractor for this requirement or a decision not to continue the current contract by award of additional term options. Since neither of those conditions has occurred at the present time, the transition plan is not in effect. The contract includes three out-year options to maintain and distribute the CC4CARB collection without new compounds being added. During this period, it is anticipated the bulk of samples will have been distributed to external researchers. Any remaining samples not too degraded to keep will be transitioned to a chemical storage facility, likely part of the Interventional Agents Chemistry Services. It is anticipated that the global need for new classes of antibacterial agents will continue beyond the end of this contract and the residual CC4CARB collection may serve as the starting point of an, as yet undetermined, chemistry initiative to continue addressing the problem in a strategically new way.

12. Program office certification.

This is to certify that the portions of this justification that have been developed by the undersigned program office personnel, including supporting information and/or data verifying the Government's minimum needs, schedule requirements and other rationale for other than full and open competition, are accurate and complete.

OFFICIAL	NAME & TITLE	SIGNATURE	DATE
Project Officer	Jeremy Starr, Program Officer; COR	Jeremy T. Starr -S 	Digitally signed by Jeremy T. Starr -S Date: 2024.09.23 11:20:57 -04'00'
Project Officer's Immediate Supervisor	Richard Sciotti, Section Chief	Richard J. Sciotti -S 	Digitally signed by Richard J. Sciotti -S Date: 2024.09.23 11:53:40 -04'00'
Head of the Sponsoring Program Office	Emily Erbelding, Director	Emily J. Erbelding -S 	Digitally signed by Emily J. Erbelding -S Date: 2024.09.23 21:30:20 -04'00'

13. Contracting Officer certification.

This is to certify that the justification for the proposed acquisition has been reviewed and that to the best of my knowledge and belief the information and/or data provided to support the rationale and recommendation for approval is accurate and complete.

OFFICIAL	NAME & TITLE	SIGNATURE	DATE
Contracting Officer	Aytaj Vily	Aytaj Vily -S Digitally signed by Aytaj Vily -S Date: 2024.09.30 11:39:51 -04'00'	

14. Chief of the Contracting Office and Head of the Contracting Activity signature(s).

OFFICIAL	NAME & TITLE	SIGNATURE	DATE
Chief of the Contracting Office	George Kennedy, Director, Office of Acquisitions, DEA, NIAID, NIH	George W. Kennedy -S Digitally signed by George W. Kennedy -S Date: 2024.10.10 14:35:19 -04'00'	
Head of the Contracting Activity	Phillip D. Osborne, <i>For</i> Deputy Director, Office of Acquisition and Logistics Management, Acting Head of Contracting Activity	TIFFANY D. CHADWICK -S Digitally signed by TIFFANY D. CHADWICK -S Date: 2024.11.15 11:53:00 -05'00'	
R&D Competition Advocate	Dr. Liza Q. Bundesen, Deputy Director Office of Extramural Research, National Institutes of Health	Liza Q. Bundesen -S Digitally signed by Liza Q. Bundesen -S Date: 2024.11.21 08:15:54 -05'00'	